

Appendix: Using Hierarchical Models to Estimate Heterogeneous Effects

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1 Replication Analysis Details

To produce Figure 1 in the manuscript, I downloaded all replication files from the Harvard Dataverse for papers in the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics*. The analysis covers from 2010 to 2024 because of lags in replication data upload and approval for 2025 and 2026. This also means that the analysis covers the year the replication data was uploaded, not the year of publication. This is important because publication often lags acceptance by a year.

I then identified code files in the replication materials, downloading scripts in R, Stata, Python and other coding languages. Within those files, I used regular expressions to identify models with an interaction, uses of associated tools such as marginal effects or STATA's margins command, and machine-learning tools for heterogeneous effects. I also parsed comments. The two lists below summarize regular expressions for interactions and machine-learning.

Interaction Patterns

- `(lm|glm|felm|feols|plm|ivreg)\s*\([^)]*\s*~\^[^)]*\s*[:]`
- `feols?\s*\([^)]*\s*|`
- `[ci]\.\w+#`
- `\w+##\w+`
- `dydx\s*\s*(`
- `margins\s*,.*at\s*\s*(`
- `margineffects|slopes\s*\s*(|comparisons\s*\s*(`
- `heterogeneous|eity)\s*(effect|treatment)?`
- `conditional\s*(average)?\s*(treatment)?\s*effect`
- `subgroup\s*analysis|stratified\s*analysis`
- `moderation|ing|or)\s*(effect|analysis|variable)?`
- `interaction\s*(effect|term|model)`
- `interplot|interact_plot|marginsplot|coefplot.*#`

ML Heterogeneity Patterns

- `causal_forest\s*\(`
- `library\s*\(\s*grf\s*\)|require\s*\(\s*grf\s*\)`
- `average_treatment_effect|predict.*causal_forest`
- `bartc\s*\(||library\s*\(\s*bartCause\s*\)`
- `bcf\s*\(||library\s*\(\s*bcf\s*\)`
- `\bCATE\b|conditional\s+average\s+treatment\s+effect`
- `\bITE\b|individual(ized)?\s+treatment\s+effect`
- `(causal|treatment)\s*(forest|tree|learning)`
- `DoubleML|double.*machine.*learning|dml_`
- `[STXR] [-_] ?learner|metalearner`

2 Simulation Data Generating Process

This section summarizes the data generating process for the simulations in the main text. Both simulations fix the sample size at $n = 2000$ and vary the number of groups, with different numbers of observations in each group.

2.1 Equal Group Size Simulation

For the first simulation, I generate k binary grouping variables, where $k \in \{3, 4, 5, 6, 7\}$:

$$x_{ij} \sim \text{Bernoulli}(0.5) \quad \text{for } j = 1, \dots, k$$

The simulation defines the groups by unique combinations of these variables, producing 2^k groups. With three modifiers, there are 8 groups, while 7 modifiers produces 128 groups.

The treatment indicator is randomly assigned, also with a bernoulli draw:

$$T_i \sim \text{Bernoulli}(0.5)$$

Three control variables predict the baseline outcome but do not modify the treatment effect:

$$z_{i1} \sim N(0, 1)$$

$$z_{i2} \sim \text{Bernoulli}(0.5)$$

$$z_{i3} \sim N(0, 1)$$

True group-specific treatment effects are drawn from a normal distribution with mean μ_τ and variance σ_τ , both of which are fixed:

$$\tau_g \sim N(\mu_\tau, \sigma_\tau) \quad \text{where } \mu_\tau = 0.2, \sigma_\tau = 0.3$$

The outcome is then generated as:

$$y_i = \beta_1 z_{i1} + \beta_2 z_{i2} + \beta_3 z_{i3} + \tau_{g[i]} T_i + \epsilon_i$$

$$\epsilon_i \sim N(0, \sigma_y) \quad \text{where } \sigma_y = 1$$

I also fix the control coefficients $\beta = (0.4, -0.3, 0.25)$.

The third simulation replaces the group treatment effects draw with a student-t distribution, using either 3 or 5 degrees of freedom.

$$\tau_g \sim \text{Student T}(df, \mu_\tau, \sigma_\tau) \quad \text{where } df = (3, 5), \mu_\tau = 0.2, \sigma_\tau = 0.3$$

2.2 Unequal Group Size Simulation

The second simulation uses the same data generating process but fixes $k = 6$ (64 groups) and varies the probability parameter for the grouping variables. Rather than fixing all probabilities at 0.5, I vary them symmetrically across groups to create imbalanced group sizes. I consider four scenarios:

1. Balanced (all probabilities at 0.5)
2. Mild imbalance (0.35, 0.4, 0.5, 0.5, 0.6, 0.65)
3. moderate imbalance (0.25, 0.35, 0.5, 0.5, 0.65, 0.7)
4. Strong imbalance (0.2, 0.3, 0.5, 0.5, 0.675, 0.75)

At maximum imbalance, some groups may have as few as one observation while others are much larger. I set the simulation so that all groups have at least one observation.

3 Tomz and Weeks Reanalysis

This section summarizes the hierarchical models of Tomz and Weeks (2021) and details the underlying R code. Both models use the nonlinear lambda specification in brms, which separates the outcome into two linked equations: a heterogeneity equation modeling the treatment effect (λ) and a control equation.

3.1 R Code: Tomz and Weeks Reanalysis

First, the following model formula expresses a heterogeneous treatments model where the impact of alliances varies with the experimental conditions (regime, stakes, costs, and region). The `treat_group` indicator marks unique combinations of these experimental factors for the group-level random intercept. The control equation includes respondent demographics and foreign policy dispositions.

```
bf(
  force ~ lambda * alliance + controls,

  lambda ~ regime + stakes + costs + region_txt + (1 | treat_group),

  controls ~ white + male + hawk + intl + dem + rep + age + ed4,

  nl = TRUE
)
```

The second formula expresses a respondent heterogeneity model where the impact of alliances varies with demographic characteristics (party, news interest, race, and gender). The `het_group` indicator marks unique combinations of these demographic modifiers. The control equation includes experimental conditions and respondent age and education.

```
bf(
  force ~ lambda * alliance + controls,

  lambda ~ (dem + rep + high_newsint + white + male) + (1 | het_group),

  controls ~ regime + stakes + costs + africa + europe + asia + age + ed4,

  nl = TRUE
)
```

3.2 Priors: Tomz and Weeks Reanalysis

Table 1 summarizes the prior distributions for both models. All priors are weakly informative relative to the scale of the data. The lambda coefficients receive a $N(0, 1)$ prior, while

control coefficients receive a tighter $N(0, 0.5)$ prior.

Parameter	Prior
λ coefficients	$N(0, 1)$
Control coefficients	$N(0, 1)$
σ (residual SD)	half- $N(0, 1)$
σ_θ (group SD)	half- $N(0, 1)$

Table 1. *Summary of priors in hierarchical models.*

3.3 *Who Responds to Alliances?*

To further explore the potential application of hierarchical models, this section examines how demographic factors modify the impact of alliances. I used party, news interest, race, and gender to modify the impact of alliances. I selected these variables because Tomz and Weeks examine party and news interest as potential modifiers, but gender (Barnhart et al., 2020) and race are also salient modifiers. I control for other experimental manipulations, age and education. Age and education are important group-level controls because they are group-level variables that are correlated across party, race and gender. Following TW's OLS analysis, I use a Gaussian likelihood, although the outcome is a binary variable.

The resulting hierarchical model thus encompasses four subgroup comparisons in TW's appendix while adding two more modifiers. All of the comparisons in the hierarchical model would require at least six pairwise interaction models. What TW present in four figures, I show with two.

To start, I present the two regressions from the hierarchical model. Figure 1 plots how different variables shape either the impact of an alliance or support for intervention. One facet of this figure gives the coefficient estimates from the model of heterogeneity, and a second gives the control estimates.

High news interest is the strongest predictor of how alliances impact support for war. Respondents with high news interest respond to alliances by .15 more than others in this survey, and that relationship could be as small as .05 or as large as .22. This differs from TW's conclusion that respondents with high and low news interest respond in roughly the same way, likely because the hierarchical model accounts for other factors that are correlated with news interest.

Other demographic predictors do not differentiate responses to alliances as clearly. The impact of alliances may be somewhat greater for white and male respondents, but those differences are not clearly different from zero as the 95% credible intervals include zero and small negative values. Similarly, Republicans and Democrats are not all that different from independents in how they respond to alliances. This matches TW's conclusion that partisanship does not create substantial differences in alliance effects. The intercept in this equation is theoretically meaningful, as it captures the impact of alliances when all the modifiers are zero. For independent, non-white women with low news interest, alliances increase support for force by roughly .15.

The control equation suggests that high stakes, high costs, and partner democracy all increase support for intervention. Region is less consequential, as neither Asia, Eastern Europe nor Africa is clearly different from the reference category of Latin America. Because neither education nor age are equal to zero, the intercept is not directly interpretable.

While it is possible to construct a rough profile of who responds most to alliances using the estimates in Figure 1, I provide a more precise summary in Figure 2. This figure plots the median estimate in each group of respondents, after grouping respondents based on race, gender, news interest and partisanship. The median depends on the regression coefficients in Figure 1, with additional noise from the partial pooling components. There are 24 unique estimates, all of which are clearly greater than zero. Alliances thus increase support for using

Determinants of Support for Force and Alliance Impact

Hierarchical Model

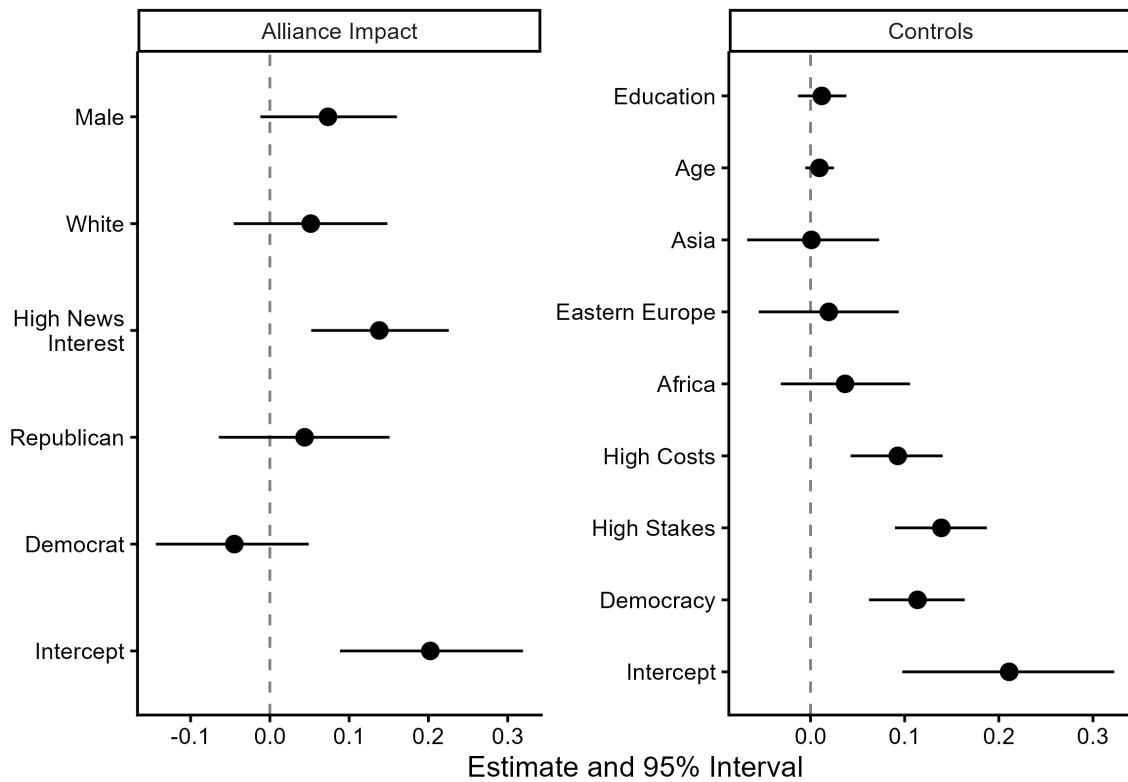


Figure 1. Regression coefficients from a hierarchical model of how an alliance shapes public support for military intervention. The left facet reports predictors of group alliance effects, and the right facet the control variables in the equation with the alliance treatment. Points give the posterior median and error bars the 95% credible interval.

Alliance Impact:
 Party, Gender, Race, and News Interest

		Low News Interest		High News Interest		
Race	White	0.23	0.29	0.34	0.42	Democrat
	Non-White	0.13	0.22	0.3	0.38	
	White	0.25	0.32	0.39	0.45	Independent
	Non-White	0.2	0.29	0.35	0.41	
	White	0.3	0.37	0.44	0.5	Republican
	Non-White	0.25	0.32	0.38	0.46	
		Female	Male	Female	Male	Gender

Figure 2. *Estimated impact of alliance on support for military intervention in different subgroups of respondents. Text gives the exact posterior median estimate.*

force among all demographic groups, but the magnitude of the effect varies.

Alliances most impact white Republican men with high interest in the news. In this group, the impact of alliances is .5, because this group was skeptical of intervention without an alliance rationale. The gap between high and low news interest in the other demographic groups is .15, which matches the coefficient in the heterogeneity equation.

The weakest impact of alliances appears for non-white, politically independent women with low news interest. Here, alliances increase support for force by 13%. White respondents respond by 8% more, Democrats by 4% more, and Republicans by 10%. The accumulation of these different effects create substantial gaps between the groups.

4 Bush and Prather Reanalysis

In the following, I again demonstrate how the model works by reanalyzing a study by Bush and Prather (2020) (BP hereafter). This study examines how foreign meddling in elections impacts support for economic engagement with the meddler. One of their experiments examines how Russian or German engagement in the 2016 US election impacts mass support for trade and investment with those countries. I use a heterogeneous effects model to check their results and further explore their findings.

BP employ a 2x2x2 factorial experiment. This design randomizes whether a foreign country is interfering in the 2016 election, the country and their attitude towards Trump and whether the potential economic ties entail greater trade or investment in the United States. In one meddling treatment, Russia expresses support for Trump and in the other, Germany expresses opposition to Trump.

BP hypothesize that individuals will prefer economic engagement with states that support their candidate. They thus examine how the impact of side-taking varies with the direction of the endorsement, individual political preferences, and the economic issue. To do this, they use four tables and figures, and rely on eight t-tests of differences in means between groups of roughly 25 respondents.

I use the same nonlinear lambda specification as in the Tomz and Weeks reanalysis. The heterogeneous effects model estimates the impact of side-taking on economic engagement, with the treatment effect (λ) modeled as a function of potential modifiers. The heterogeneity equation includes a dummy indicator of intention to vote for Clinton, an indicator of whether the experiment deals with trade or investment, female gender, and high political engagement. I measure political engagement as a sum of political knowledge and interest, dichotomized at the median. Groups are defined by unique combinations of these four binary modifiers, producing 16 groups.

The outcome variable is a scale from one to four that measures support for greater economic engagement with the potential partner. To approximate BP's analyses, I use a normal outcome likelihood. The control equation includes college education and employment status.

4.1 R Code: Bush and Prather Reanalysis

The following model formula expresses the heterogeneous effects model:

```
bf(
  ipe_support ~ lambda * treat_germrus + controls,

  lambda ~ (w2_vote_hill + woman + high_pol_engage + invest_cond) +
    (1 | het_group),

  controls ~ college_educ + employ_dum + woman + high_pol_engage

  nl = TRUE
)
```

The priors follow the same structure as the Tomz and Weeks models, with $N(0, 1)$ on coefficients in the heterogeneity equation and $N(0, 0.5)$ on control coefficients.

These estimates corroborate BP's findings. Clinton voters are more likely to support economic engagement with Germany when Germany has opposed Donald Trump. This has the largest impact on attitudes towards trade among highly engaged respondents. Low engagement Clinton voters are less responsive.

Trump voters are less responsive to German opposition to Trump. The effects are mostly null, and only for engaged voters in the investment condition does the endorsement have a clearly non-zero effect. This positive effect may reflect residual opposition to Trump in some sections of the Republican party.

References

- Barnhart, Joslyn N, Robert F Trager, Elizabeth N Saunders and Allan Dafoe. 2020. "The Suffragist Peace." *International Organization* 74(4):633–670.
- Bush, Sarah Sunn and Lauren Prather. 2020. "Foreign Meddling and Mass Attitudes Toward International Economic Engagement." *International Organization* 74(2):584–609.
- Tomz, Michael and Jessica L.P. Weeks. 2021. "Military Alliances and Public Support for War." *International Studies Quarterly* 65(3):811–824.

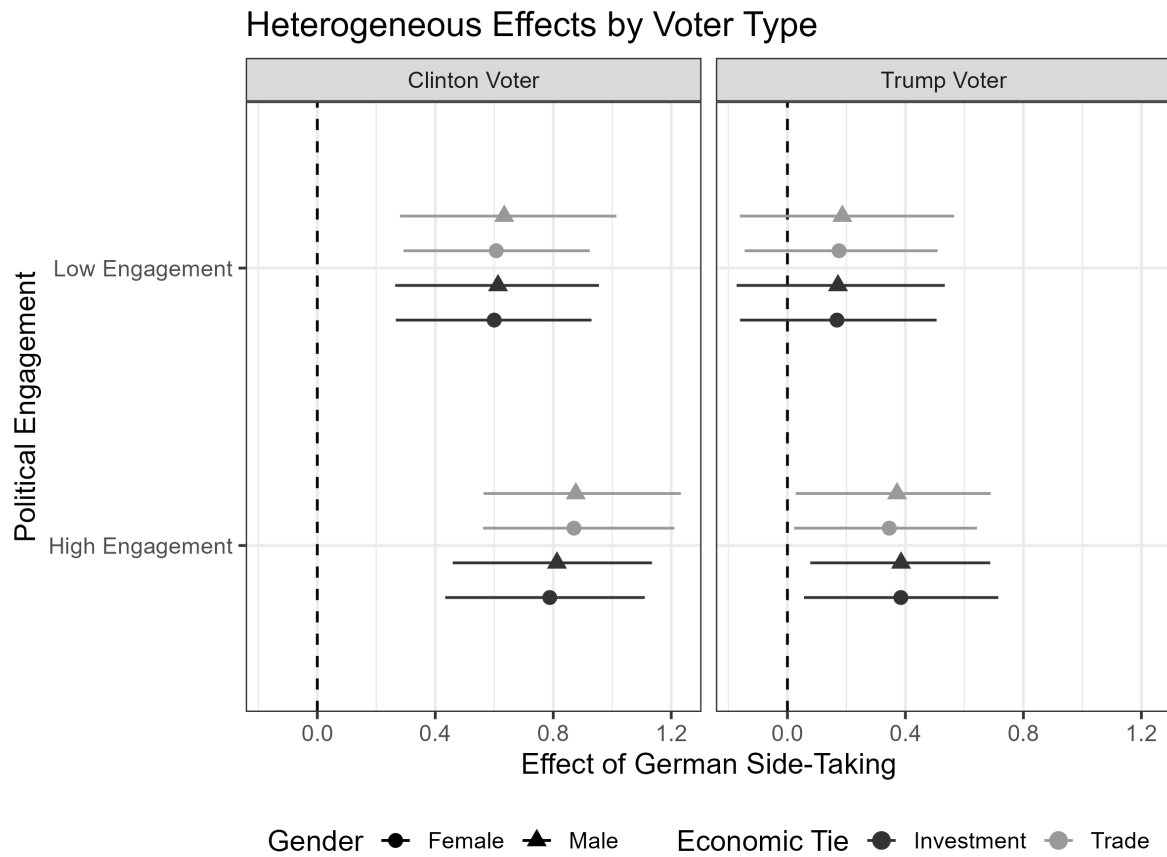


Figure 3. *Heterogeneous effects of German side-taking in US elections on support for economic engagement with Germany. Each estimate reflects a treated group with a unique combination of other treatments and demographic characteristics.*